

Unified Architecture Specification

MONOLITHIC SYSTEM COMPONENT INTERACTION DIAGRAM

PRESENTATION LAYER (CLIENT BROWSER)

HTML5 Document

Unified glassmorphic client interface structure. Contains Sign-In and Workspace overlays.

JS Logic Coordinator

app.js manages view switching, form fetches, localized sessionStorage caching, and click triggers.

CSS3 Style Tokens

Defines responsive flex/grid layouts, theme color tokens, and tagline animation keyframes.

API CONTROLLER LAYER (SPRING MVC REST)

UserController

Handles user session verification, credential sign-in, sign-up, and recovery updates.

TaskController

Coordinates REST endpoints for task creation, column state transitions, and dependency maps.

ReviewController

Handles audit submissions, approvals, rejections, and fallback overrides.

IncidentController

Manages logs for production alarms and updates active incident lifecycles.

SERVICE LAYER (CORE BUSINESS RULES ENGINE)

TaskService (Cycle Guard)

Applies DFS validation to block circular dependencies and handles state cascades when dependencies complete.

ReviewService (Auditor Engine)

Handles reviewer allocations and initiates automatic fallback assignments if an auditor is out-of-office.

NotificationService (Deadline Scan)

Triggers daily cron scanners to discover upcoming task deadlines and dispatches system alerts.

PERSISTENCE & DATABASE LAYER (JPA / SQL)

Spring Data JPA Repositories

Hibernate ORM handles mapping data transactions to SQL schemas recursively.

Embedded H2 Database

In-memory relational schema seeded instantly with data.sql for rapid testing and zero-setup deployment.

-  Client Presentation
-  REST Request Handler
-  Business Logic Service
-  Data Persistence Mapping